

IELTS Vocabulary Audio App

Design Document v1.0

A hands-free system for recalling English while parenting



User

Personal use



Platform

iPhone / iPad



Type

PWA (Web App)



Goal

IELTS 7.0+

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Where to use AI vs. where not to

01 Project Overview

Background & Requirements



No time to read vocab books while parenting



Want to study by listening during housework



Have a physical word book (200+ words) & MP4 audio



Two audio modes: EN→JP and JP→EN



Track whether words are memorized + review



Quiz on collocations (prepositions like 'in')



Personal use only on iPhone/iPad

Constraints



Zero server cost (free tier only)



Dev in spare time (15-20 min/day with infant)



Dev environment: Linux Ubuntu



Playback device: iPhone / iPad



Core Design Philosophy

Not "digitize a word book" but

A hands-free recall system

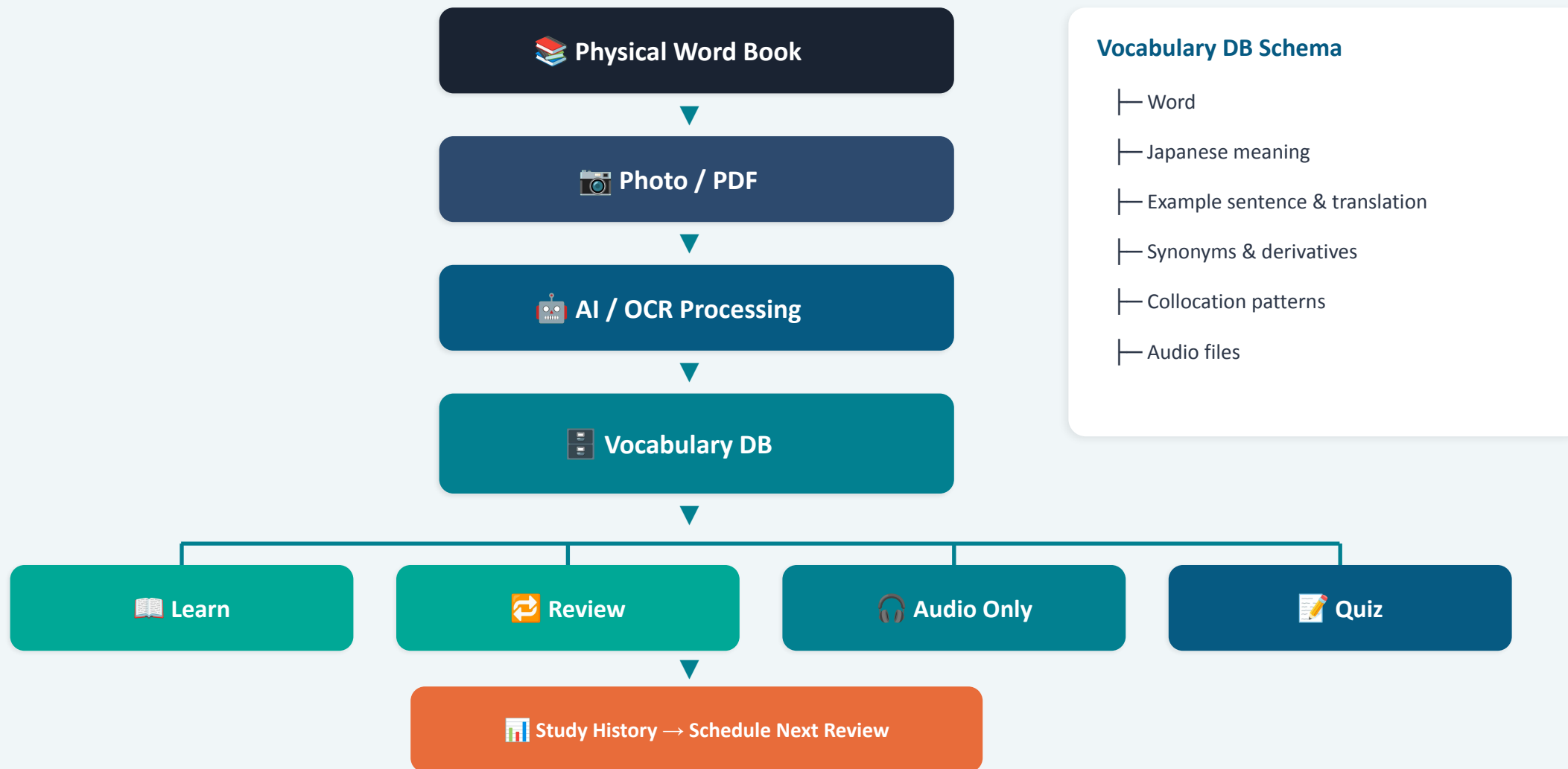
designed for busy parents

Recall → Answer structure

Using silence gaps to trigger active recall

02 System Architecture

Separate data from learning methods — add features without rebuilding data



03 Data Model Design

Design around "learning items", not just words — decompose each word into multiple things to learn

Word Object

id	"001"
level	"5"
word	"interested"
meaning	"to have interest"
example	"I'm interested in..."
example_translation	"Japanese translation..."
synonyms[]	["curious", ...]
derivatives[]	["interest", ...]
patterns[]	["interested in"]
audio	audio file path

Pattern Object (separated)

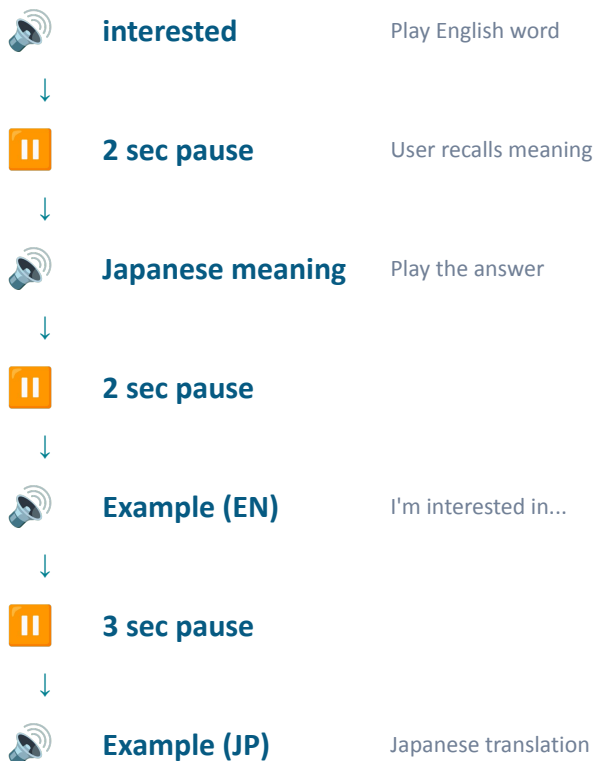
- └─ pattern → "interested in"
- └─ meaning → "to be interested in ~"
- └─ example → example sentence
- └─ audio → audio file

learning_status (multi-axis)

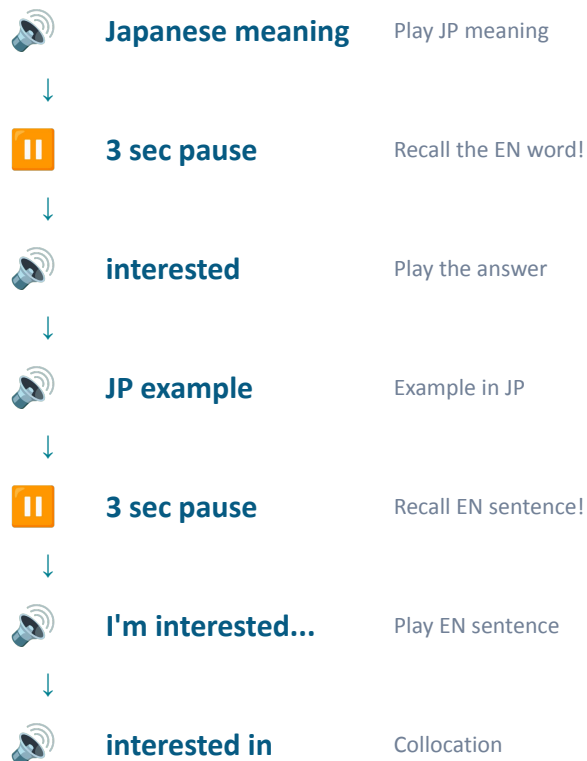
EN → JP	meaning_en_to_ja	◎ / △ / ×
JP → EN	meaning_ja_to_en	◎ / △ / ×
Sentence	example	◎ / △ / ×
Collocation	pattern	◎ / △ / ×
Spelling	spelling	◎ / △ / ×

04 Learning Engine — Audio Modes

Mode A: English → Japanese



Mode B: Japanese → English



Hands-Free Study Mode

Fully audio, no screen needed
The most valuable feature for parents

Select "15-min session", then:

1. Play Japanese meaning
2. 3 sec pause (recall time)
3. Play English word (answer)
4. Play English example
5. 3 sec pause
6. Play JP translation
7. Play collocation pattern
8. → Next word

*Session summary after 15 min
(reviewed, new, weak words)*

04b Learn Mode vs Review Mode

Learn = Input (absorb information) / Review = Retrieval (practice recalling)

Learn Mode (Input)

Walk through all info for a new word

1. Show / play the English word
2. Confirm pronunciation
3. Japanese meaning
4. Example sentence (English)
5. Example translation (Japanese)
6. Collocation (e.g. interested in)
7. Synonyms
8. Derivatives

Works with screen or audio-only

Review Mode (Retrieval)

Minimize hints, focus on active recall

 Japanese meaning	Prompt given
 3 sec pause	User recalls
 "interested"	Answer revealed
 "interested in"	Collocation
 Example sentence	Context check

Self-evaluation after each:

🎯 Instant recall = 3 pts

⚠️ Needed a moment = 1 pt

× Couldn't recall = 0 pts

05 Quiz Engine Design

Auto-generate 8 quiz types from data — quiz engine is separated from learning data

Type 1

English → Japanese

e.g.:
interested → ?

Type 2

Japanese → English

e.g.:
meaning → ?

Type 3

Fill-in (preposition)

e.g.:
interested ____ science

Type 4

Sentence → Word

e.g.:
I'm ____ in science.

Type 5

JP Sentence → EN

e.g.:
JP sentence → EN

Type 6

Collocation

e.g.:
interested ____

Type 7

Synonym Match

e.g.:
closest to interested?

Type 8

Derivative

e.g.:
interest → interested

06 Progress Tracking

Multi-Axis Evaluation

Skill	⦿ Instant	△ Slow	× Missed
EN→JP	3	—	—
JP→EN	—	1	—
Sentence	3	—	—
Collocation	—	—	0
Spelling	—	1	—

→ "Knows 'interested' but weak on 'interested in'"

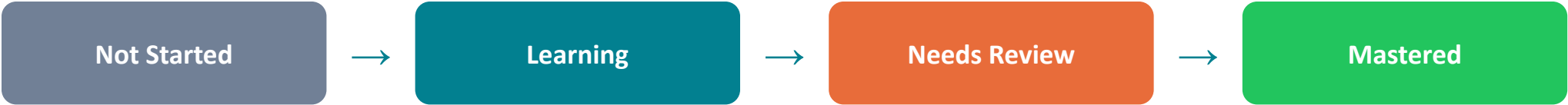
Review Algorithm (Spaced Repetition)

Start simple:

- × Couldn't recall → Tomorrow
- △ Needed a moment → In 3 days
- ⦿ Instant recall → In 7 days
- ⦿ 2x in a row → In 14 days
- ⦿ 3x in a row → In 30 days

Can evolve into Anki-style SRS later
Don't over-engineer from the start

Learning Status Transitions



07 Data Pipeline

The hardest part isn't the app — it's turning a physical book into structured data



Step 1

Photograph

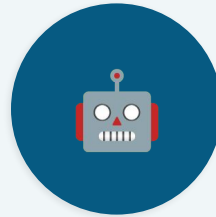
Take photos of
the word book



Step 2

OCR

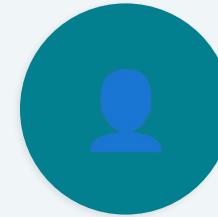
Tesseract.js
text recognition



Step 3

AI Structuring

Claude API
outputs JSON



Step 4

Human Review

Manual fix of
OCR errors



Step 5

App Import

Import JSON
into the app

Step 3 Output Example (JSON)

```
{
  "id": "001",
  "level": "5",
  "word": "interested",
  "meaning": "to have interest in",
  "example": "I'm interested in learning English.",
  "example_translation": "Japanese translation...",
  "synonyms": ["curious"],
  "derivatives": ["interest"],
  "patterns": ["interested in"]
}
```

Audio File Management

Separate word_audio and example_audio

audio/

001_word.mp3 <- TTS generated

001_example.mp4 <- Existing MP4

002_word.mp3

002_example.mp4

Missing word audio -> generate via TTS

09 Screen Design (Wireframes)

My Vocabulary

Today's Review
23 words

Learn New
10 words

Audio Study

Quiz

Progress

Home Screen

Study Card

interested
to have interest

I'm interested in
learning English.

Play Next

Instant

Slow

Missed

Study Card Screen

Session Complete Screen

Today's 15 Minutes

Words studied30

Reviewed20

New words10

Weak words:

interested in
responsible for

UX Key Insight

Not "study for 15 min" but
"play this during 15 min of chores"

10 Tech Stack

Phase 1: PWA → upgrade to native iOS only if needed / No App Store required



Frontend

React + TypeScript

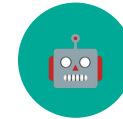
Mobile-responsive
PWA support



OCR

Tesseract.js

Client-side processing
Free & offline



AI Structuring

Claude API

User's own API key
~\$0.5-2/month



TTS

Web Speech API

Browser-native
Completely free



Hosting

Vercel (free tier)

Zero server cost
Auto-deploy



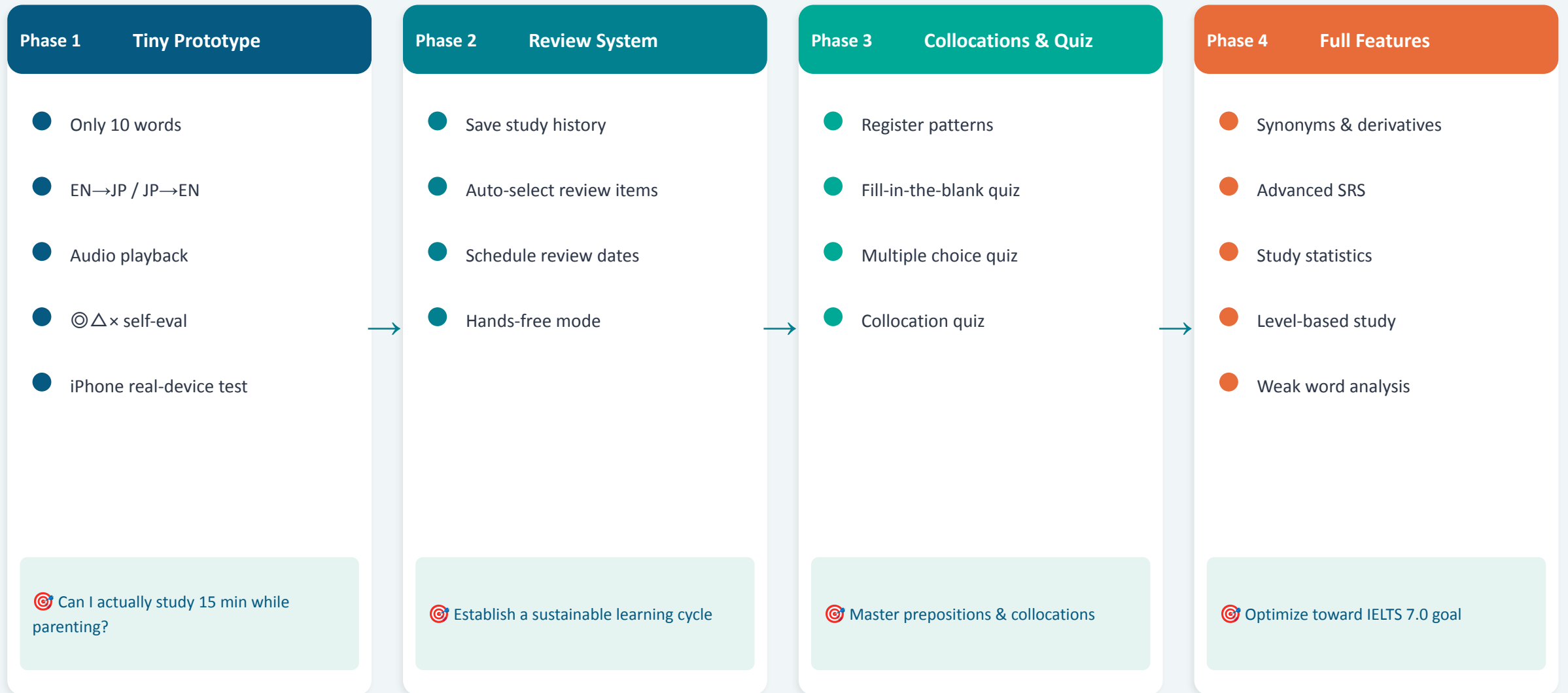
State Mgmt

React Hooks

localStorage persistence
No auth needed

⚠ Browser background audio has limitations → test on real iPhone early

11 Development Phases



Don't build a beautiful app first. Build 10 words on iPhone for 15 min of hands-free study.

12 AI Strategy & Summary

Use AI for data preparation, not as a runtime dependency in the app



AI Handles (Pre-processing)

- PDF → OCR text extraction
- Structuring word data (JSON)
- Collocation pattern extraction
- Synonym & derivative organization
- Quiz question auto-generation
- Cloze test generation from examples



App Handles (Runtime)

- Audio playback & background play
- Study history (localStorage)
- Review schedule management
- Correct / incorrect tracking
- Data display & UI rendering
- Session management (15-min timer)

Portfolio Value

Not just "I built a vocabulary app" but a case study in designing a Learning UX that enables active recall under parenting constraints, with minimal cognitive load.

A project at the intersection of UX, AI, and Learning Design.